

PUDASJARVI, Finland

WMO: 028670

Lat: 65.367N

Lon: 27.017E

Elev: 117

StdP: 99.93

Time Zone: 2.00 (EUE)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.7	-30.9	0.2	-28.1	-28.2	0.3	-25.6	10.1	-2.6	8.4	-0.4	0.5	160	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.3	16.4	22.5	15.8	20.7	14.9	17.6	21.8	16.6	20.7	15.8	19.7	2.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.5	18.9	15.0	10.8	18.1	14.1	10.2	17.0	49.9	21.4	47.0	21.1	44.5	19.7	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.7	6.0	-29.8	26.8	2.1	2.2	-31.3	28.4	-32.5	29.7	-33.7	30.9	-35.2	32.5		
(5)			-30.2	19.4	2.0	1.5	-31.7	20.4	-32.9	21.3	-34.0	22.1	-35.5	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-10.5	-10.3	-5.1	-0.3	5.9	12.8	14.9	12.8	7.0	1.0	-5.4	-7.7	(6)
(7)		DBStd	10.36	7.71	7.91	5.05	4.95	3.96	3.64	2.92	3.20	3.63	5.19	5.77	7.32	(7)
(8)		HDD10.0	3537	636	568	468	308	139	14	1	9	103	280	462	548	(8)
(9)		HDD18.3	6212	894	801	727	558	384	170	111	172	338	538	712	807	(9)
(10)		CDD10.0	375	0	0	0	0	13	97	153	97	14	0	0	0	(10)
(11)		CDD18.3	10	0	0	0	0	0	3	5	2	0	0	0	0	(11)
(12)		CDH23.3	94	0	0	0	0	3	47	33	12	0	0	0	0	(12)
(13)		CDH26.7	8	0	0	0	0	0	4	4	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.6	2.4	2.5	2.9	2.7	2.8	2.6	2.4	2.6	2.8	2.5	2.5	(14)	
(15)	Precipitation	PrecAvg	647	46	39	35	31	53	65	85	71	59	57	55	52	(15)
(16)		PrecMax	875	88	71	55	59	92	125	139	143	114	113	96	91	(16)
(17)		PrecMin	503	14	8	13	9	10	22	28	23	4	11	14	14	(17)
(18)		PrecStd	91	19	18	12	15	19	25	28	30	27	25	22	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	4.2	5.0	14.7	22.8	27.0	26.5	24.8	18.7	11.3	4.9	3.6	(19)
(20)			MCWB	1.7	2.3	2.3	8.4	13.5	16.7	19.6	17.6	14.9	9.9	4.2	2.2	(20)
(21)		2%	DB	1.4	2.6	3.6	11.1	18.9	24.7	23.9	22.7	16.1	9.8	3.6	2.3	(21)
(22)			MCWB	0.4	1.5	1.4	6.3	12.4	15.5	17.0	16.3	12.4	8.7	3.0	1.5	(22)
(23)		5%	DB	0.4	1.3	2.3	8.9	16.0	22.3	22.2	20.9	14.2	8.6	2.7	1.6	(23)
(24)			MCWB	-0.2	0.6	0.7	5.0	10.3	14.9	16.1	15.6	12.0	7.7	2.1	1.0	(24)
(25)		10%	DB	-1.0	0.3	1.2	6.7	13.1	20.0	20.5	19.0	12.5	7.4	1.6	0.7	(25)
(26)			MCWB	-1.5	-0.3	0.0	3.7	8.2	13.9	15.2	15.0	10.7	6.6	1.1	0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.8	3.1	8.9	15.0	18.1	19.9	18.5	15.2	10.3	4.7	2.7	(27)
(28)			MCDWB	2.9	3.5	4.7	14.0	18.7	23.4	26.1	22.3	17.8	10.7	4.9	3.2	(28)
(29)		2%	WB	0.7	1.7	1.7	6.7	12.5	16.6	17.8	17.2	13.7	9.0	3.4	1.7	(29)
(30)			MCDWB	1.3	2.4	2.8	10.7	18.3	22.0	21.8	21.2	15.4	9.6	3.7	2.0	(30)
(31)		5%	WB	-0.1	0.7	1.0	5.3	10.4	15.7	16.7	16.2	12.2	7.8	2.3	1.0	(31)
(32)			MCDWB	0.3	1.3	2.1	8.2	15.1	21.4	20.5	20.0	13.6	8.4	2.7	1.5	(32)
(33)		10%	WB	-1.4	-0.3	0.2	4.0	8.7	14.5	16.0	15.3	10.9	6.6	1.1	0.3	(33)
(34)			MCDWB	-1.0	0.4	1.1	6.6	12.3	19.2	19.5	18.6	12.3	7.3	1.6	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	6.7	7.8	8.3	8.4	8.5	7.9	7.9	6.6	5.0	5.3	6.1	(35)
(36)			MCDBR	5.8	4.7	7.5	10.3	13.7	12.9	11.3	11.3	8.2	5.8	3.6	4.4	(36)
(37)			MCWBR	5.6	4.5	6.0	6.6	7.5	5.6	5.5	6.0	5.4	4.9	3.7	4.4	(37)
(38)		5% WB	MCDBR	5.2	4.4	5.8	9.7	12.5	10.5	9.2	9.8	6.8	5.6	3.7	4.5	(38)
(39)			MCWBR	5.1	4.3	4.9	6.3	7.0	5.7	5.0	5.8	4.8	5.2	3.9	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.201	0.272	0.302	0.338	0.343	0.344	0.364	0.345	0.321	0.297	0.347	N/A	(40)	
(41)		taud	1.938	2.109	2.163	2.253	2.454	2.472	2.435	2.471	2.524	2.369	2.619	N/A	(41)	
(42)		Ebn at Noon	280	596	760	825	865	870	837	820	763	587	269	N/A	(42)	
(43)		Edh at Noon	21	58	88	104	94	95	96	84	64	46	20	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.11	0.60	2.02	3.71	4.71	5.20	4.89	3.60	1.95	0.69	0.15	0.03	(44)	
(45)		RadStd	0.01	0.11	0.20	0.34	0.46	0.41	0.49	0.35	0.18	0.11	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page